Chapter:6

Objectives

Preparation of Solutions - Normal Solution and Normal Saline

By the end of this practical, students should be able to:

1. Define a solution.
2. Differentiate between various types of solutions.
3. Explain the properties of solutes, solvents, and solutions.
4. Demonstrate the calculation and preparation of solutions of different concentrations.

Introduction

Solution:
A homogeneous mixture or single-phase mixture of molecules, atoms, or ions of two or more components is called a solution.

- **Solute:** The substance that dissolves in a solvent is called a solute.
- **Solvent:** A substance that has the ability to disperse a solute is called a solvent.

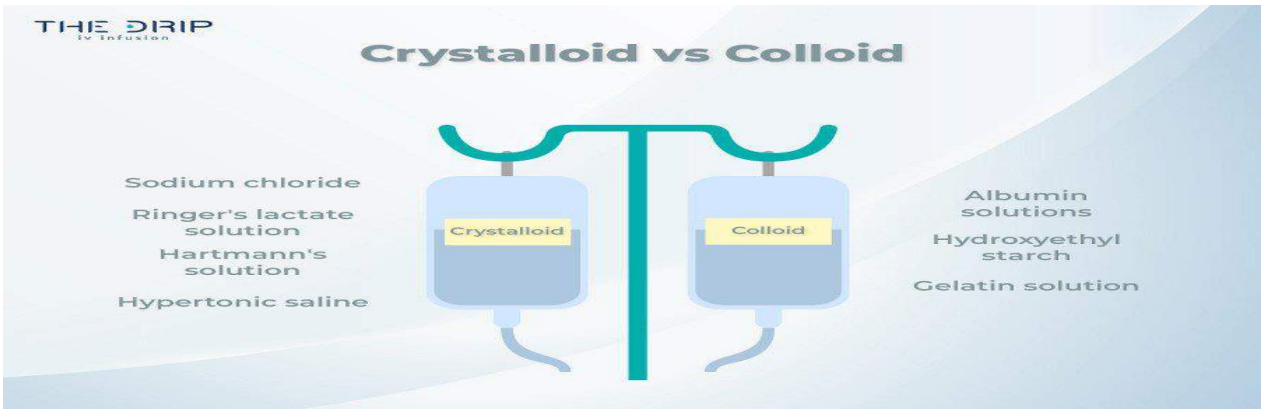
Types of Solutions

There are three main types of solutions based on the size of solute particles:

1. **True Solution:**
   A solution in which the particle size of the solute is less than 1 nm (nanometer).
   Example: Salt solution.
2. **Colloidal Solution:**
   A solution in which the particle size of the solute ranges from 1 to 200 nm. The particles are dispersed in the solution.
   Example: Protein and polysaccharide solutions.



3. **Crystalloid Solutions:**
   A solution of low molecular weight ions that can pass through a semi-permeable membrane.
   Example: Sodium chloride solution.



Difference Between Emulsion/Suspension and Their Pharmaceutical Applications

Emulsion:
An emulsion is a solution of two immiscible liquids.
Example: Oil in water emulsion.

Suspension:
A suspension is a solution in which the particle size of the solute is greater than 200 micrometers and the particles are suspended in the solution, visible with the naked eye.
Example: Mucaine suspension.

Pharmaceutical Applications:

- **Emulsions:**
  1. Taste and smell of the oil are masked.
  2. The oil becomes more easily digested.
  3. There is uniform distribution of the drug.
- **Suspensions:**
  1. The disagreeable taste of the drug can be masked (e.g., Chloramphenicol palmitate).
  2. Suspensions are generally more stable than solutions.
  3. An ideal dosage form for patients who have difficulty swallowing tablets or capsules (e.g., infants, children, elderly patients).
  4. Drugs are absorbed more rapidly from a suspension than from tablets or capsules.



Preparation of Different Solutions

1. **W/V% Solution:**
   A weight per unit volume solution (W/V%) means the number of grams of solute per 100 mL of solution.
   Example: A 5% W/V solution means 5 grams of solute in 100 mL of solution.
2. **W/W% Solution:**
   A weight per unit weight solution (W/W%) means the number of grams of solute per 100 grams of solution.
   Example: A 5% W/W solution means 5 grams of solute in 95 grams of solvent.
3. **V/V% Solution:**
   A volume per unit volume solution (V/V%) means the number of millilitres of solute per 100 mL of solution.
   Example: A 5% V/V solution means 5 mL of solute in 95 mL of solvent.

Atomic Weight, Molecular Weight, Equivalent Weight, Mole, and Osmole

1. **Atomic Weight:**
   The sum of the number of protons and neutrons in the nucleus of an atom.
2. **Molecular Weight:**
   The sum of atomic weights of atoms presents in a molecule.
3. **Equivalent Weight:**
   The gram molecular weight divided by the acidity, basicity, or valency of the substance.
4. **Mole:**
   The gram molecular weight of a substance.
5. **Osmole:**
   Any substance, molecule, or ion that can exert osmotic pressure.

Molarity, Molality, Normality, Osmolality, and Osmolarity

1. **Molarity (M):**
   The number of moles of a solute in 1 litre (1000 mL) of solution.
2. **Normality (N):**
   The gram equivalent weight of a solute in 1 litre (1000 mL) of solution.
3. **Osmolality:**
   The number of particles of solute per unit weight of water, irrespective of the size and nature of the particles. Units are mmol/Kg of water.
4. **Osmolarity:**
   The number of particles of solute per litre of solution. Units are mmol/L.

Tonicity: Isotonic, Hypertonic, and Hypotonic Solutions

1. **Isotonic Solution**:
   A solution that has the **same osmolarity** as the cell's intracellular fluid.
   No net movement of water occurs across the cell membrane.
   Example: **0.9% NaCl** (Normal saline)
2. **Hypertonic Solution**:
   A solution that has a **higher osmolarity** than the cell's interior.
   Water moves **out of the cell**, causing the cell to **shrink** .
   Example: **3% NaCl**
3. **Hypotonic Solution**:
   A solution that has a **lower osmolarity** than the cell's interior.
   Water moves **into the cell**, causing the cell to **swell** and possibly **burst** (lysis).
   Example: **0.45% NaCl**

Practical 1: Preparation of a 5% W/V Solution of Glucose

Objective:
To prepare a 5% W/V solution of glucose in distilled water.

Principle:
A W/V% solution means the weight (in grams) of a solute in 100 mL of solution. Glucose is a hydrophilic (polar) molecule that easily dissolves in water. A 5% glucose solution is prepared by dissolving 5 grams of glucose in 100 mL of distilled water.

Requirements:

1. Glucose
2. Distilled water
3. Pipette
4. Flask
5. Weighing machine

Procedure:

1. Weigh 5 grams of glucose using the weighing machine.
2. Take a small amount of distilled water using the pipette and add it to a flask.
3. Add the weighed glucose to the flask and mix the two contents until a homogeneous solution is formed.
4. Adjust the volume of the solution to 100 mL using distilled water.

Result:
5 grams of glucose dissolved in distilled water to make a 5% W/V glucose solution.

Interpretation:
A 5% glucose solution is not only used for chemical purposes but also has clinical importance. Glucose 5% IV infusion BP is used for treating carbohydrate and fluid depletion. It can also serve as a vehicle and diluent for compatible medicinal products for parenteral administration. The concentration of glucose may vary depending on the patient's condition.

Practical 2: Preparation of a 0.9% Solution of Sodium Chloride (NaCl)

Objective:
To prepare a 0.9% NaCl solution (Normal Saline) in distilled water.

Principle:
A W/V% solution means the weight (in grams) of a solute in 100 mL of solution. Sodium chloride (NaCl) is a hydrophilic (polar) molecule that dissolves easily in water. A 0.9% NaCl solution is prepared by dissolving 0.9 grams of sodium chloride in 100 mL of distilled water.

Requirements:

1. Sodium chloride (NaCl)
2. Distilled water
3. Pipette
4. Flask
5. Weighing machine

Procedure:

1. Weigh 0.9 grams of sodium chloride using the weighing machine.
2. Take a small amount of distilled water using the pipette and add it to a flask.
3. Add the weighed sodium chloride to the flask and mix the contents until a homogeneous solution is achieved.
4. Adjust the volume of the solution to 100 mL using distilled water.

Result:
0.9 grams of sodium chloride dissolved in distilled water to make a 0.9% NaCl solution.

Interpretation:
The 0.9% sodium chloride solution, commonly known as physiological saline, is used for chemical and clinical purposes. It is often used for treating isotonic extracellular dehydration, sodium depletion, and as a vehicle or diluent for compatible drugs in parenteral administration. Different concentrations of NaCl may be used depending on the patient's condition.